

Italy — Monthly Intelligence Report v4.2

Edition 005 — The Puglia Corridor

Where the energy transition is outrunning the grid. edition · July 2026 · SSI v4.2 (refreshed June 2026)

CONTENTS

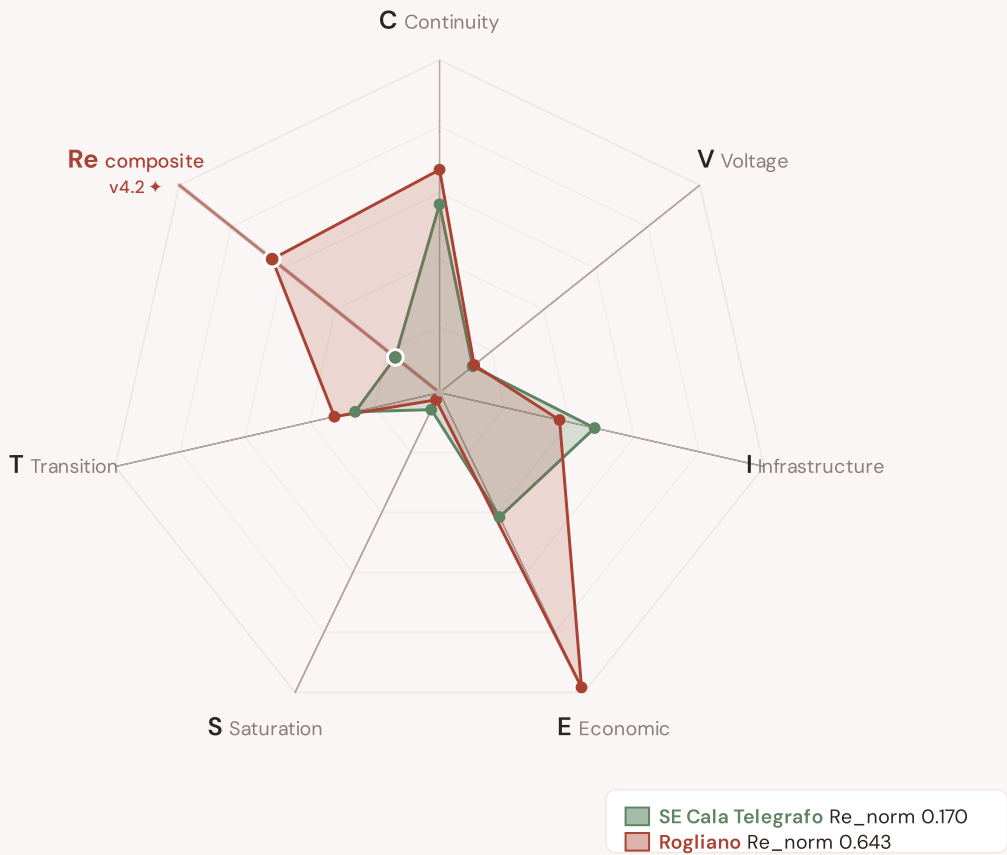
A Executive Summary	E European Context Card
B Cyber & Economic Exposure Monitor	F SSI Enhanced Neural Network
C Data Refresh & Changelog	F.5 Anti-maladaptation Gate (Radar 2 · W1–W10) v4.2
D Deep-Dive: The Puglia Corridor	G Looking Ahead

SECTION A

Executive Summary

Radar 1 — v4.2: now 7-axis with the new Re composite

v4.0.2 Radar 1 had 6 axes (C/V/I/E/S/T). **v4.2 adds Re — the bounded resilience composite** capturing the new modifier family (R6c flood + R6d wildfire + R6e winter + R8 adapt + R9 compound + R10 just) in one normalised scalar (Re_norm ∈ [0, 1] via clip((Re_raw − 0.920) / (1.787 − 0.920), 0, 1)). Two worked examples overlaid: **SE Cala Telegrafo** (low-exposure reference, Re_norm 0.170) · **Rogliano** (high-exposure reference, Re_norm 0.643). The Re axis is colour-highlighted in cayenne to emphasise the v4.2 architectural addition.



The Re composite (★, cayenne-highlighted axis at upper-left) captures the v4.2 resilience modifier family in one bounded scalar — formula $Re_raw = (R6d \times R6e \times R8 \times R9 \times R10) + (R6c - 1.00)$, bounds [0.920, 1.787]. Per Convention #6 §2.3.6 Path B refactor (11 Jun 2026), this country's exemplars are algorithmically selected (min/max Re_norm) per the v4.2 methodology. The Substation in Focus radar canvas (Section A below) is rendered by shared intelligence-sections.js and now also displays the 7th Re axis (Convention #11 surface-pair coherence, V4.2-disc-24 closure 12 Jun 2026).

SUBSTATIONS SCORED

4,293

475 EHV · 3,035 HV · 783 distribution-tier

GRID LINES MAPPED

14,221

~46,000 km coverage

MC SIMULATIONS

42.9 M

10,000 per substation

PUBLIC DATA SOURCES

34

101 variables · zero proprietary

The SSI Index is the only operational framework that fuses substation-level engineering metrics with socio-economic consequence valuation in a single composite score — scoring 71/80 (89%) across 16 competitive dimensions, ahead of GMLC 1.1 (48/80), EPRI (40/80), and academic MCDA frameworks (41/80). Its engine processes 146,000+ source data points per run, executes 42.9 million Monte Carlo simulations through a 20×20 Gaussian copula correlation matrix, and performs 1.78 billion arithmetic operations — producing 244,701 output data points with full uncertainty quantification across 4,293 substations and 14,221 grid lines spanning ~46,000 km. v4.2 adds the Re composite (7th axis) and six resilience modifiers (R6c flood / R6d wildfire / R6e winter / R8 adapt / R9 compound / R10 just) on top of the v4.0.2 foundation.

Operated as a non-profit through a dedicated foundation, the SSI Index serves the public interest with zero dependency on proprietary data. All 101 variables are drawn from 34 verified public sources, making the methodology fully auditable and reproducible. Initially covering Italy’s entire transmission and distribution grid, the Index is progressively being rolled out to all OECD countries — applying the same silo-breaking approach: fusing grid risk with societal exposure at asset-level resolution wherever open data permits.

Substation in Focus

Roquette Italia Cassano Spinola

Genova, Liguria · 132 kV

0.418

Medium

0.108

41th

R_FINAL

CLASSIFICATION

CI WIDTH

FLEET PERCENTILE

Re composite v4.2 7th axis

Re_raw 1.1123 · Re_norm 0.222

LOW EXPOSURE

COMPONENTS

C

Continuity: 0.253 / 0.30 (84%)

I

Infrastructure: 0.443 / 0.25 (177%)

S

Saturation: 0.059 / 0.20 (30%)

V

Voltage: 0.140 / 0.10 (140%)

E

Economic: 0.362 / 0.10 (362%)

T

Transition: 0.366 / 0.05 (732%)

MODIFIERS

R3

Consequence: 1.172 ↑ amplifies

R4

Graph Criticality: 1.050 ↑ amplifies

R6a

Restoration: 1.068 ↑ amplifies

R6b

Seismic: 1.025 ↑ amplifies

R7

Digital Readiness: 1.003 → neutral

This month's spotlight — automatically selected for July 2026 — is **Roquette Italia Cassano Spinola** in Genova, Liguria. With an R_final of 0.418 (Medium band, 41th fleet percentile), its risk profile is dominated by the T (Transition) component at 732% of its weight ceiling, followed by E (Economic) at 362%. Modifiers collectively shift R_base by 56.8%, reflecting the substation's specific geographic, socio-economic, and network context.

SECTION B — RECURRING MODULE

Cyber & Economic Exposure Monitor

Why this section exists: Traditional grid risk frameworks treat cybersecurity and economic vulnerability as separate domains. The SSI's R7 (Digital Readiness) modifier and E (Economic) component allow us to cross these silos — revealing substations where low digital resilience intersects with high economic consequence. This is precisely the anticipatory intelligence that PNRR digitalisation planning requires.

CYBER HIGH EXPOSURE

669

15.6% of fleet

BLIND SPOTS

1229

High/Critical + R7 < median

AVG R7 MODIFIER

1.0009

Range [0.99, 1.05]

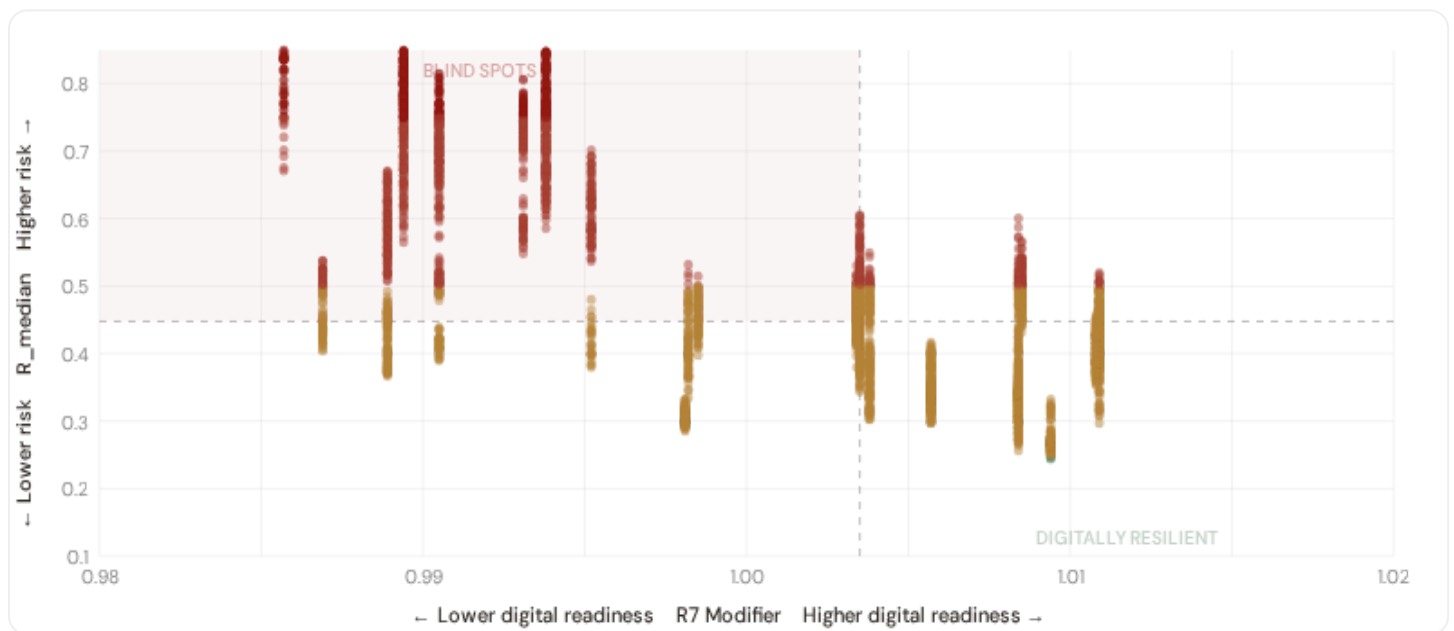
HIGH-CONSEQUENCE SUBS

1,056

24.6% · R3 ≥ 1.16

B.1 Cyber-Physical Risk Matrix

Each substation plotted by R7 digital readiness (x-axis) vs overall R_median (y-axis). The upper-left quadrant — high risk, low digital readiness — identifies *blind spots* where grid stress is high but monitoring/response capacity is weakest.



● Low ● Medium ● High ● Critical

Dashed lines = fleet medians

The cyber-physical matrix identifies **1229 blind-spot substations** — scoring High or Critical on overall risk while sitting below the fleet median for digital readiness ($R7 < 1.0035$). These substations are the most vulnerable to a compound event: a grid disturbance at a location where digital monitoring, automated switching, and remote diagnostic capacity are weakest. The blind spots concentrate in **Sicilia (308)**, **Campania (231)**, **Puglia (187)**. Across the full fleet, 669 substations (15.6%) carry a HIGH cyber-exposure classification, while 207 (4.8%) are in the LOW tier.

B.2 Economic Impact by Business Fabric

Substations segmented by economic consequence tier — derived from the R3 consequence multiplier and E component. Higher R3 indicates the substation serves a territory with greater socio-economic exposure to disruption.

 **Capital-Intensive / High-Consequence**

Est. VoLL: €15–30/kWh (EUR)

Capital nodes — highest economic loss per interruption (semiconductor fabs, data centers, refineries, hospitals)

1,056 substations (24.6%) High/Critical: **253** (24.0%) Avg R: **0.454** Avg E: **0.399** / 0.10

Low 0

Med 803

High 213

Crit 40

 **Industrial / Medium-Large Enterprise**

Est. VoLL: €8–15/kWh (EUR)

Industrial belt — significant continuity requirements (manufacturing, logistics hubs)

1,067 substations (24.9%) High/Critical: **266** (24.9%) Avg R: **0.460** Avg E: **0.419** / 0.10

Low 0

Med 801

High 215

Crit 51

 **Commercial / SME-Dense**

Est. VoLL: €3–8/kWh (EUR)

Service economy + retail — moderate individual impact (commercial districts, mixed-use)

1,083 substations (25.2%) High/Critical: **295** (27.2%) Avg R: **0.470** Avg E: **0.436** / 0.10

Low 0

Med 788

High 233

Crit 62

 **Light / Agricultural / Rural**

Est. VoLL: €1–3/kWh (EUR)

Sparse activity, agricultural land — lower VoLL (residential, agricultural)

1,087 substations (25.3%) High/Critical: **622** (57.2%) Avg R: **0.578** Avg E: **0.612** / 0.10

Low 4

Med 461

High 321

Crit 301

Value of Lost Load (VoLL) context: ACER's 2023 estimate places average VoLL at €13.1/kWh for industrial customers and €3.2/kWh for residential. The SSI's R3 consequence multiplier allows us to identify which substations serve the most economically exposed territories. **253 substations** combine capital-intensive economic fabric (R3 ≥ 1.16) with High or Critical risk classification — representing the highest VoLL-weighted exposure in the fleet.

The capital-intensive tier concentrates in **Lombardia (218), Toscana (108), Emilia-Romagna (101)** — regions where industrial corridors depend on uninterrupted power for continuous processes. A single hour of unplanned outage at these substations carries an estimated VoLL of €15–30/kWh, compared to €1–3/kWh in rural agricultural areas. The fleet-wide average energy poverty rate is 10.0%, but this masks sharp regional variation: Southern regions combine higher energy poverty with higher grid risk, creating a double vulnerability that the E component captures. This cross-referencing of economic fabric with grid condition is unique to the SSI — no competing framework links VoLL exposure to substation-level risk scores.

B.3 Province Digital Readiness Ranking

Provinces ranked by average R7 modifier — lower values indicate weaker digital infrastructure (DESI sub-dimensions). Cross-referenced with average R_{median} to identify provinces combining digital exposure with high grid stress.



WORST DIGITAL READINESS — TOP 15 REGION REGIONS

Longer bar = higher R7 = better digital infrastructure.

1	Vibo Valentia ▲ high R		0.9857
2	Cosenza ▲ high R		0.9857
3	Reggio Calabria ▲ high R		0.9857
4	Crotone ▲ high R		0.9857
5	Catanzaro ▲ high R		0.9857
6	Matera ▲ high R		0.9869
7	Potenza ▲ high R		0.9869
8	Isernia		0.9889
9	Campobasso ▲ high R		0.9889
10	Messina ▲ high R		0.9894
11	Agrigento ▲ high R		0.9894
12	Enna ▲ high R		0.9894
13	Palermo ▲ high R		0.9894
14	Catania ▲ high R		0.9894
15	Ragusa ▲ high R		0.9894

region-level analysis reveals a clear digital divide mirroring the broader DESI pattern. **Vibo Valentia** has the lowest average R7 (0.9857), while **Parma** leads at 1.0109 — a spread of 0.0252 across the R7 range. **43 region regions** combine below-median digital readiness with above-median grid risk, making them priority targets for digitalisation investment. The R7 modifier is intentionally narrow to reflect the current limitations of open DESI data — as NIS2 compliance reporting matures, future editions will expand R7’s dynamic range and incorporate operational technology (OT) security indicators.

B.4 Monthly Pulse

Inaugural edition — Baseline Snapshot (July 2026). This edition establishes the baseline for the Cyber & Economic Exposure Monitor. Key reference points: fleet average R7 = 1.0009, median = 1.0035; 669 substations classified HIGH cyber-exposure; 1229 blind-spot substations (High/Critical risk + below-median R7); 454 Critical-band substations with below-median digital readiness. Future editions will track deltas against this baseline.

SECTION C

Data Refresh & Changelog

The SSI v4.0.2 draws on **34 verified public data sources** feeding 101 variables across 6 components and 11 modifiers. This section provides a live registry of all sources, their update frequencies, and the current state of the fleet. Transparency in data vintage is central to the SSI's credibility.

TOTAL SOURCES	HOURLY — MONTHLY	QUARTERLY / ANNUAL
34	5	22

101 variables

High-frequency feeds

Regular refresh cycle

STATIC / DERIVED

8

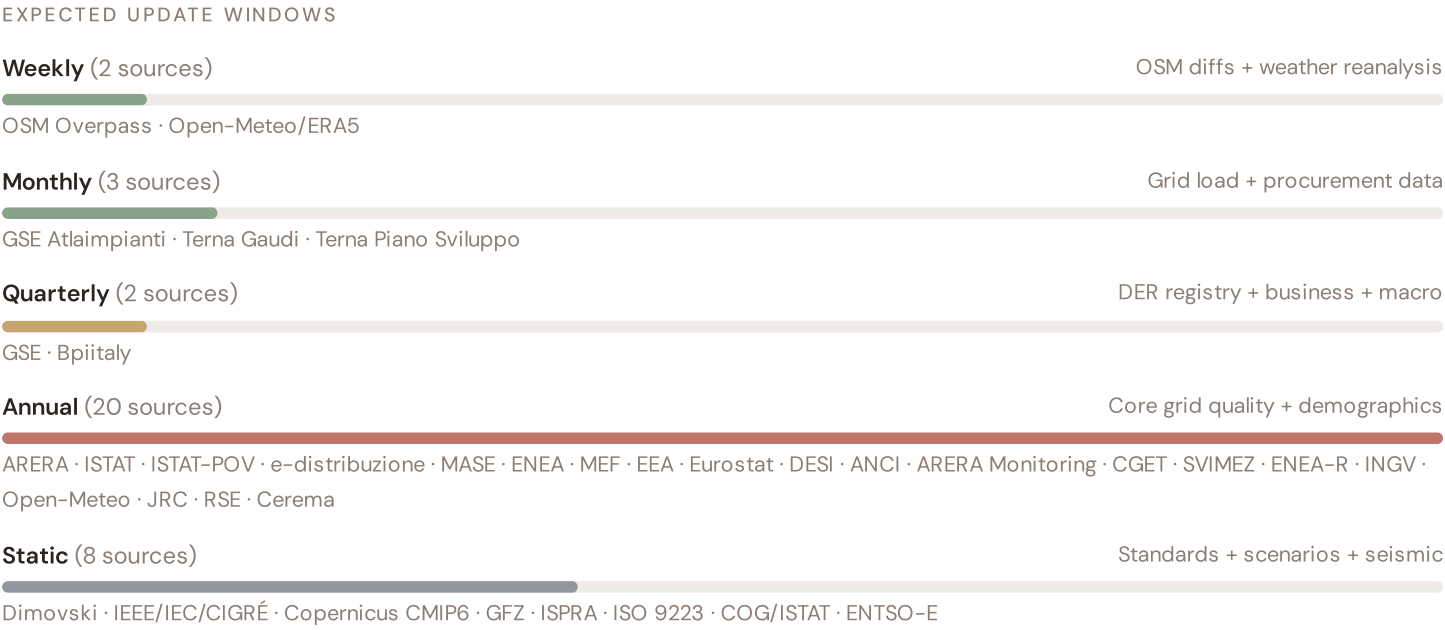
Standards + scenarios

Data Source Registry — July 2026

OSM	OSM Power Infrastructure	WEEKLY	Node/edge	3 vars
AM	Aeronautica Militare — Meteo	DAILY	~1 km	3 vars
TERNA	Terna SpA (TSO)	MONTHLY	Control area / zone	8 vars
INGV	INGV — Geofisica e Vulcanologia	CONTINUOUS	~5 km	4 vars
PCM-DPC	Dipartimento Protezione Civile	CONTINUOUS	Comune	2 vars
COPER-EM	Copernicus Emergency Management	CONTINUOUS	Event-based	1 var
CSIRT	CSIRT-Italia (ACN sub-org)	CONTINUOUS	National	1 var
GSE	GSE Gestore Servizi Energetici	QUARTERLY	Regione	4 vars
BdI	Banca d'Italia	QUARTERLY	Regione	3 vars
ENEL-XR	e-distribuzione (Enel DSO)	QUARTERLY	Provincia	3 vars
ARERA	ARERA Regulator	ANNUAL	DSO + Provincia	8 vars
ISTAT	ISTAT — Italian National Statistics	ANNUAL	Regione + Provincia ...	9 vars
ISPRA	ISPRA Environmental Protection	ANNUAL	Regione + bacino	4 vars
ACN	ACN Cybersecurity Agency	ANNUAL	National + sector	2 vars
DESI	DESI Digital Economy Index	ANNUAL	EU + NUTS-2	2 vars
EEA	EEA Air Quality e-Reporting	ANNUAL	~1 km + station	3 vars
EURO	Eurostat Energy Statistics	ANNUAL	NUTS-2	3 vars
SNAM	SNAM Gas Network Operator	ANNUAL	National	1 var
RSE	RSE Ricerca sul Sistema Energetico	ANNUAL	Regione	2 vars
MIT	MIT — Ministero Infrastrutture	ANNUAL	National	1 var
AUTBAC	Autorità di Bacino Distrettuali	ANNUAL	7 bacini	2 vars
JRC	JRC DSO Observatory	ANNUAL	DSO-level	2 vars
ARETI	Areti (ACEA Group, Roma DSO)	ANNUAL	Comuni Roma	2 vars

UNARETI	Unareti (A2A Group, Milano DSO)	ANNUAL	Milano + Brescia	2 vars
INRETE	Inrete (Hera Group, ER DSO)	ANNUAL	Emilia-Romagna	2 vars
SETDIS	Set Distribuzione (Trentino)	ANNUAL	Trentino-AA	1 var
ISPRA-INV	ISPRA IFFI Landslide Inventory	ANNUAL	Comune	1 var
ANAC	ANAC Anti-Corruption Authority	ANNUAL	National	1 var
D10	Copernicus CDS / ERA5	STATIC	0.25° (~25 km)	4 vars
IEEE-1	IEEE C57.91 / IEC 60076	STATIC	Asset-level	16 vars
CIGRE	CIGRE TB 761 Asset Management	STATIC	Asset-level	5 vars
GFZ	GFZ German Research Centre	STATIC	~5 km	2 vars
ISO-9223	ISO 9223 Corrosion	DERIVED	Regione + coastal	1 var
GME	GME Gestore dei Mercati Energetici	HOURLY	7 bidding zones	3 vars
ENTSE	ENTSO-E Transparency	HOURLY	Bidding zone	2 vars

C.1 Update Frequency Timeline



C.2 Fleet Score Snapshot

FLEET SIZE

4,293

MEDIAN R

0.448

HIGH + CRITICAL

1436

substations scored

P5=0.298 · P95=0.826

33.4% of fleet

HIGH CONFIDENCE

99%

4248 subs · CI/R < 0.50

BAND DISTRIBUTION



No primary data sources have been updated since the v4.0.2 launch baseline. All **4,293 substation scores remain unchanged** this edition. The fleet median R of 0.448 (mean 0.491) reflects a distribution skewed toward the lower bands, with 0.1% in Low and 66.5% in Medium. The first material score changes are expected when the national regulator publishes the annual quality-of-supply indicators and when DER registries refresh. High-frequency sources (OSM, weather) update weekly but feed only infrastructure topology (R4) and thermal proxy (I5), which have minimal impact on fleet-level score distribution.

C.3 Changelog — v3.4 → v4.0.2 → v4.2

v4.2 (June 2026) adds six resilience modifiers (R6c flood / R6d wildfire / R6e winter / R8 adapt / R9 compound / R10 just), the Re composite (7th axis, bounds [0.920, 1.787]), four new data sources (ISPRA IdroGEO PGRA, ISPRA SIAB wildfire, ECMWF ERA5 Bourgouin, ARERA bulk smart meter), six new variables, and W1–W10 anti-maladaptation governance gate (5/1/1/3 4-tier mesh resolution per Convention #6 §5.6) on top of the v4.0.2 baseline.

27 changes tracked across PO, P1, and P2 releases

V42-1	NEW	R6c_flood — PGRA flood vulnerability modifier (ISPRA IdroGEO, additive cliff +0.00..+0.30, P3=high/P2=medium/P1=low per Italian PGRA convention)	§5
V42-2	NEW	R6d_wildfire — Canadian Fire Weather Index modifier (EFFIS FWI + NASA FIRMS + ISPRA SIAB national-tier, multiplicative [1.00, 1.20])	§5
V42-3	NEW	R6e_winter — Bourgouin freezing-rain p99 tail-event amplifier (ERA5, multiplicative [1.00, 1.15], Italy collapses to ~1.005 per brief §6 methodologically-defensible absence)	§5
V42-4	NEW	R8_adapt — Adaptive capacity (ARERA bulk SM + ENEA + DESI, multiplicative)	§5
V42-5	NEW	R9_compound — Pairwise hazard coupling (R6b×R6c, R6c×R6d, R6c×R6e, multiplicative)	§5
V42-6	NEW	R10_just — Energy justice (EU-SILC + OIPE energy-poverty, multiplicative)	§5
V42-7	NEW	Re composite — Resilience composite (R6d × R6e × R8 × R9 × R10) + (R6c - 1.00); bounds [0.920, 1.787]	§5b
V42-8	DATA	101 variables (was 95) — +6 modifier inputs from new resilience family	§8
V42-9	DATA	34 data sources (was 30) — +4: ISPRA IdroGEO PGRA, ISPRA SIAB+Carabinieri, ECMWF ERA5 Bourgouin, ARERA bulk SM	§12

V42-10	NEW	W1-W10 5/1/1/3 4-tier audit-state split (5 Published / 1 Published-baseline ▲ / 1 Baseline-only / 3 Commercial) — mesh resolution per Convention #6 §5.6	W1-W10
F1	NEW	New T component — Energy Transition Exposure (T1)	\$2, \$4
F2	NEW	New R6a — Restoration Speed Modifier (CAIDI-based)	\$5
F3	ENHANCED	R3 enhanced — Energy Poverty Vulnerability (V_socio)	\$5
F4	ENHANCED	R4 enhanced — Graph-theoretic betweenness + bridge detection	\$5
F5	ENHANCED	R2 enhanced — Climate Trajectory (CMIP6 SSP2-4.5)	\$5
F6	ENHANCED	R7 Digital Readiness — Provincia-level DESI model	\$5
L1	NEW	New I5, I7-I9 metrics — thermal, load, corrosion, hydrogeo	\$2, \$8
L2	ENHANCED	E2 Innovation Enrichment — HRST + startup density	\$6
L3	ENHANCED	V_socio Fiscal Enrichment — MEF + ANCI + energy price	\$5
L4	ENHANCED	R3 Demographic Shift Amplifier — ISTAT net migration	\$5
L5	DATA	95/95 variables operational (v4.0.2 baseline; 101/101 in v4.2). 30 data sources total (v4.0.2; 34 in v4.2).	\$8, \$12
G1	DATA	d02 GSE Atlaimpianti upgraded to quarterly registry — Provincia-level DER registry	\$12
G2	DATA	d05 ISPRA upgraded to live API — Provincia-level geological data	\$12
G3	DATA	d06 OSM upgraded to Overpass API — 7,898 real substations	\$12
G4	NEW	R6b Network Topology modifier — centrality + ring analysis from OSM graph	\$5
G5	NEW	Network & Topology layer (H): 7 variables — Terna Piano di Sviluppo, ring analysis	\$12
G6	NEW	Output Scores layer (I): 7 variables — risk_score, ETTC, stationary probs	\$12

SECTION D — MONTHLY DEEP-DIVE

The Puglia Corridor: Where the Energy Transition Is Outrunning the Grid

Deep-dive rotation: Each edition features a substantive thematic analysis. The 12-month rotation: **Ed. 001** Puglia corridor · **002** DER saturation hotspots · **003** Climate vulnerability under CMIP6 · **004** North-South economic impact disparity · **005** Voltage quality and industrial output · **006** Infrastructure age vs investment · **007** Stress-Resilience quadrant trends · **008** T-component forward projections · **009** Provincial Grid Health Cards · **010** Academic validation and peer feedback · **011** European replication feasibility · **012** Year in review.

D.1 Why Puglia, Why Now

PUGLIA SUBSTATIONS

106

92 EHV · 14 HV

HIGH BAND

10 + 96

100.0% of corridor

CORRIDOR MEDIAN R

0.885

Fleet median: 0.448

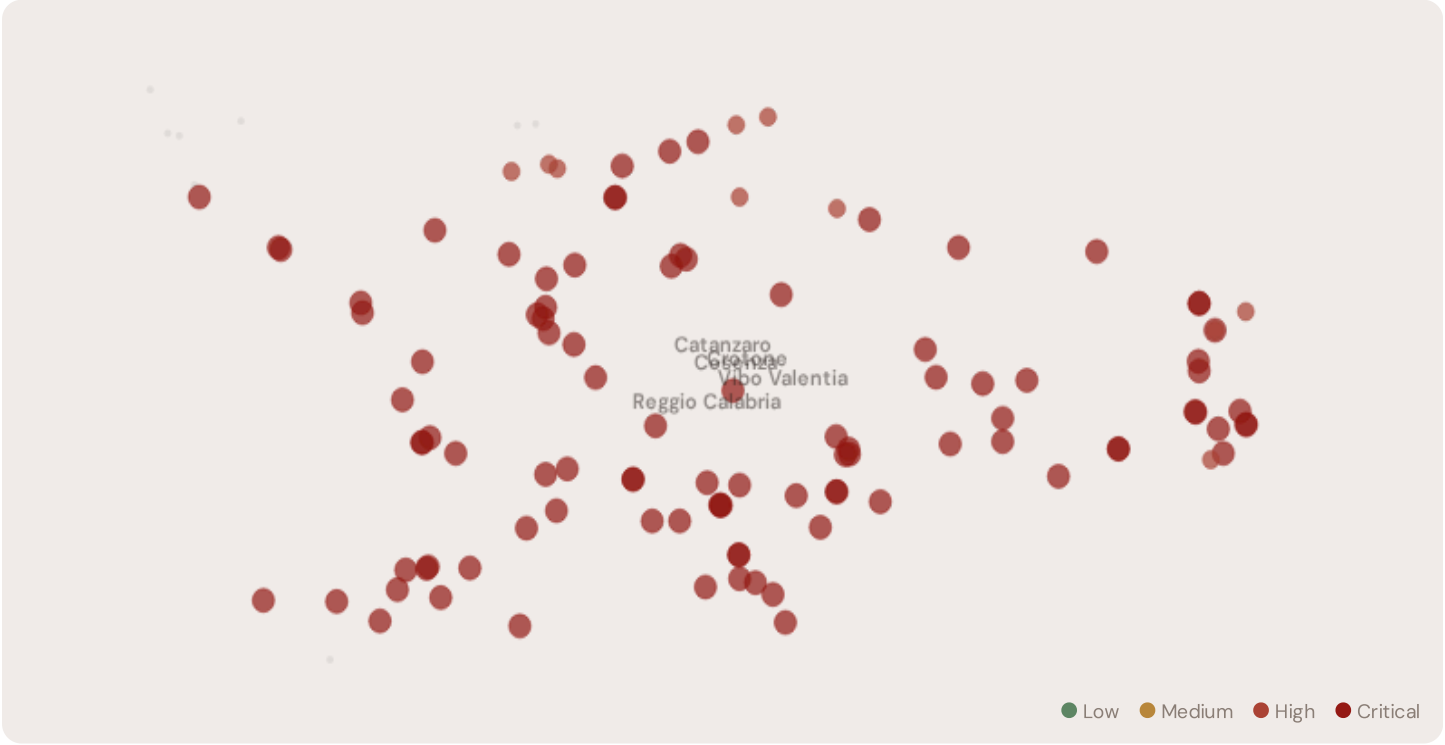
WORST PROVINCE

Vibo Valentia

Avg R: 0.932 · 18 substations

The Calabria hosts **106 substations** across 5 region regions, making it the locus of this edition's deep-dive analysis. Its risk profile is concentrated in the upper bands: **10 substations (9.4%)** are classified High and **96** Critical, with only **0** in the Low band. 100% of corridor substations score above the national fleet median of 0.448. The corridor median R of 0.885 reflects the local risk profile. The corridor concentrates the country's industrial heartland. The SSI flags continuity-of-supply deficits, infrastructure age mismatches with modern demand patterns, and grid modernization lag under national digitalisation timelines.

D.2 Corridor Map and Scoring

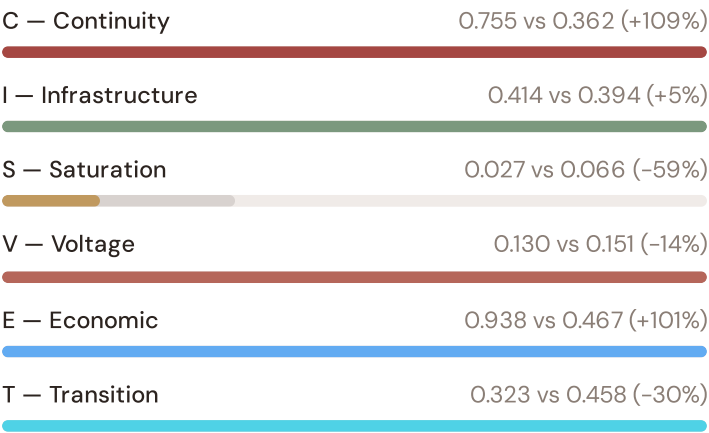


SUBSTATION	PROVINCE	R_FINAL	BAND	C	V	I	E	S	T	RE v4.2	ALERT
Rende	Vibo Valentia	0.981	Critical	1.000	0.131	0.375	0.944	0.029	0.320	C, E	
Albi	Vibo Valentia	0.980	Critical	0.974	0.130	0.338	0.909	0.027	0.356	C, E	

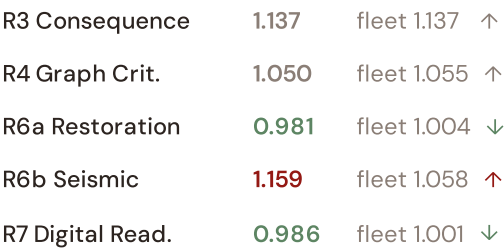
SUBSTATION	PROVINCE	R_FINAL	BAND	C	V	I	E	S	T	RE V4.2	ALERT
Magisano	Vibo Valentia	0.979	Critical	1.000	0.128	0.351	0.853	0.028	0.304	C, E	
Paola	Vibo Valentia	0.971	Critical	1.000	0.130	0.392	0.988	0.027	0.342	C, E	
Marcellinara	Vibo Valentia	0.964	Critical	0.899	0.128	0.335	0.931	0.026	0.308	C, E	
Lamezia Terme FS	Vibo Valentia	0.964	Critical	0.921	0.124	0.358	0.901	0.029	0.338	C, E	
Italc Vibo	Vibo Valentia	0.962	Critical	1.000	0.133	0.342	0.944	0.029	0.320	C, E	
Feroleto CP	Catanzaro	0.957	Critical	0.840	0.143	0.389	0.971	0.026	0.306	C, E	
Feroleto	Crotone	0.954	Critical	0.764	0.138	0.493	0.896	0.026	0.338	C, E	
Cariati	Vibo Valentia	0.954	Critical	0.960	0.129	0.376	0.890	0.029	0.298	C, E	
... 96 additional substations — explore full corridor in Digital Twin →											

D.3 Component Analysis

COMPONENT AVERAGES — CALABRIA VS FLEET



MODIFIER AVERAGES — CALABRIA VS FLEET



The dominant risk driver across the Calabria corridor is **E (Economic)**, averaging 0.938 against a fleet average of 0.467 — occupying 938% of its weight ceiling (w = 0.10). The second contributor is **T (Transition)** at 0.323 vs fleet 0.458 (646% of ceiling). Among modifiers, the R3 consequence multiplier averages 1.137 (fleet: 1.137), reflecting the socio-economic exposure of the corridor. The R6b seismic modifier averages 1.159, capturing the local seismic exposure — a dimension invisible to every competing framework.

D.4 Province Breakdown

Longer bar = lower R = better resilience.



The region regions-level breakdown reveals significant intra-regional variation. **Vibo Valentia** leads with a mean R of 0.932 across 18 substations, while **Cosenza** scores 0.807 — a spread of 0.125 within a single corridor. The SSI's substation-level granularity reveals that the Calabria is not uniformly stressed but contains distinct sub-corridors of concentrated risk — precisely the kind of spatial pattern that investment planning requires.

D.5 Implications

106 Calabria substations score $R \geq 0.650$, placing them in the upper quartile of national risk. For NEPN / national digital plan / RRP investment prioritisation, this analysis identifies the Vibo Valentia sub-corridor as the highest-priority target — where DER deployment is accelerating against ageing infrastructure with above-average continuity deficits and significant socio-economic exposure. The SSI's silo-breaking approach reveals what no single-discipline framework can: that these substations matter not only because of their engineering condition, but because the communities they serve are disproportionately vulnerable to disruption. This is precisely the anticipatory grid planning capability that the SSI was built to provide.

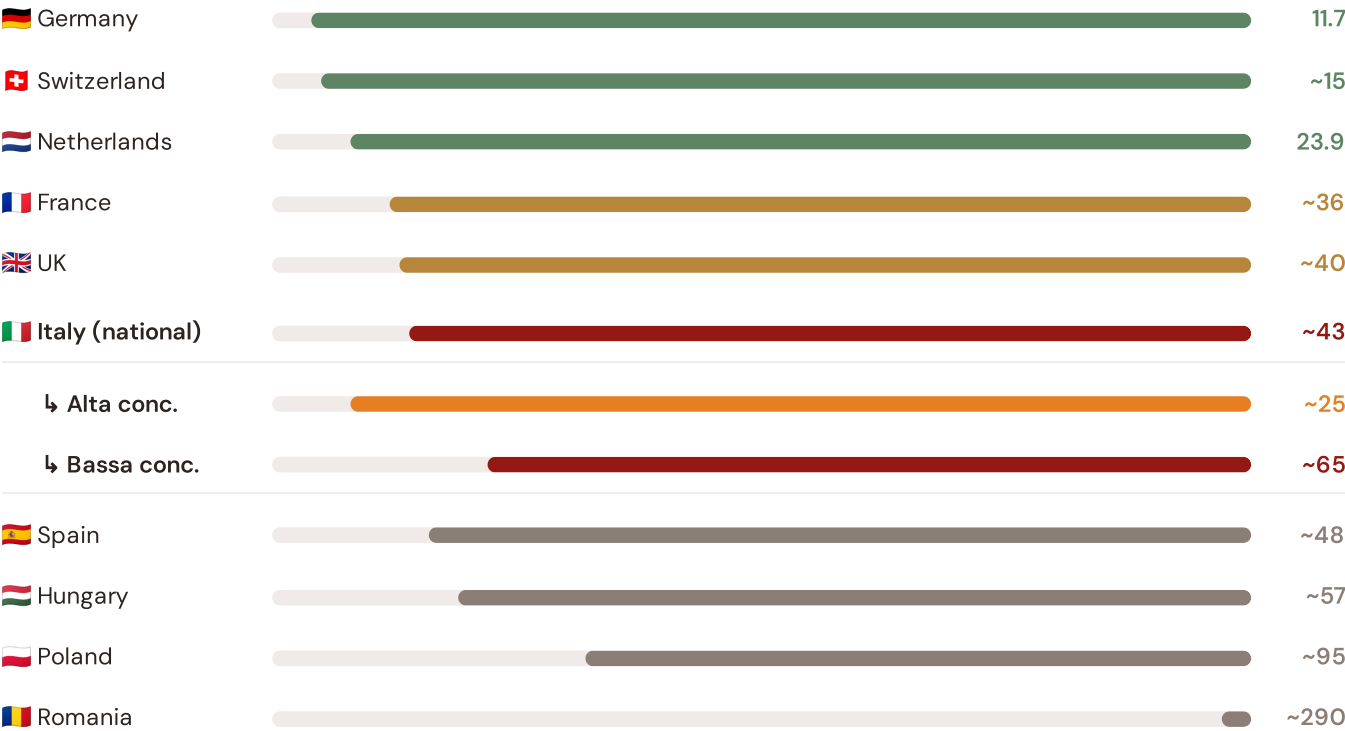
SECTION E — RECURRING MODULE

European Context Card

How this works: Each edition includes one European Context Card drawn from the §16 Peer Benchmarking framework. The card is selected to complement the month's deep-dive theme. This month: **SAIDI comparison** — because the Puglia corridor analysis hinges on Italy's continuity performance relative to European peers. Future editions rotate through: VoLL comparison (Ed. 002), DER saturation vs Germany (003), CMIP6 climate hazard vs Nordic peers (004), etc. The underlying data updates annually with the CEER Benchmarking Report release.

Unplanned SAIDI (min/customer/year) — Selected Countries

Longer bar = lower SAIDI = better reliability. Values in min/customer/year.



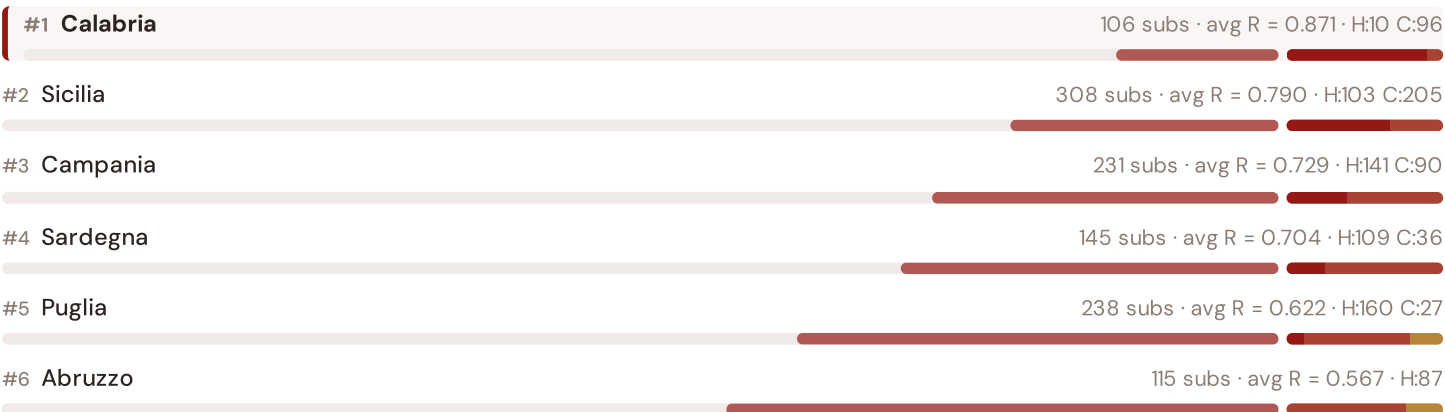
Source: CEER 7th Benchmarking Report (2022); Bundesnetzagentur (2024); ARERA (2023). Italy sub-categories from ARERA territory classification. · See §16 Peer Benchmarking for full methodology and caveats. · Updated annually with CEER BR release.

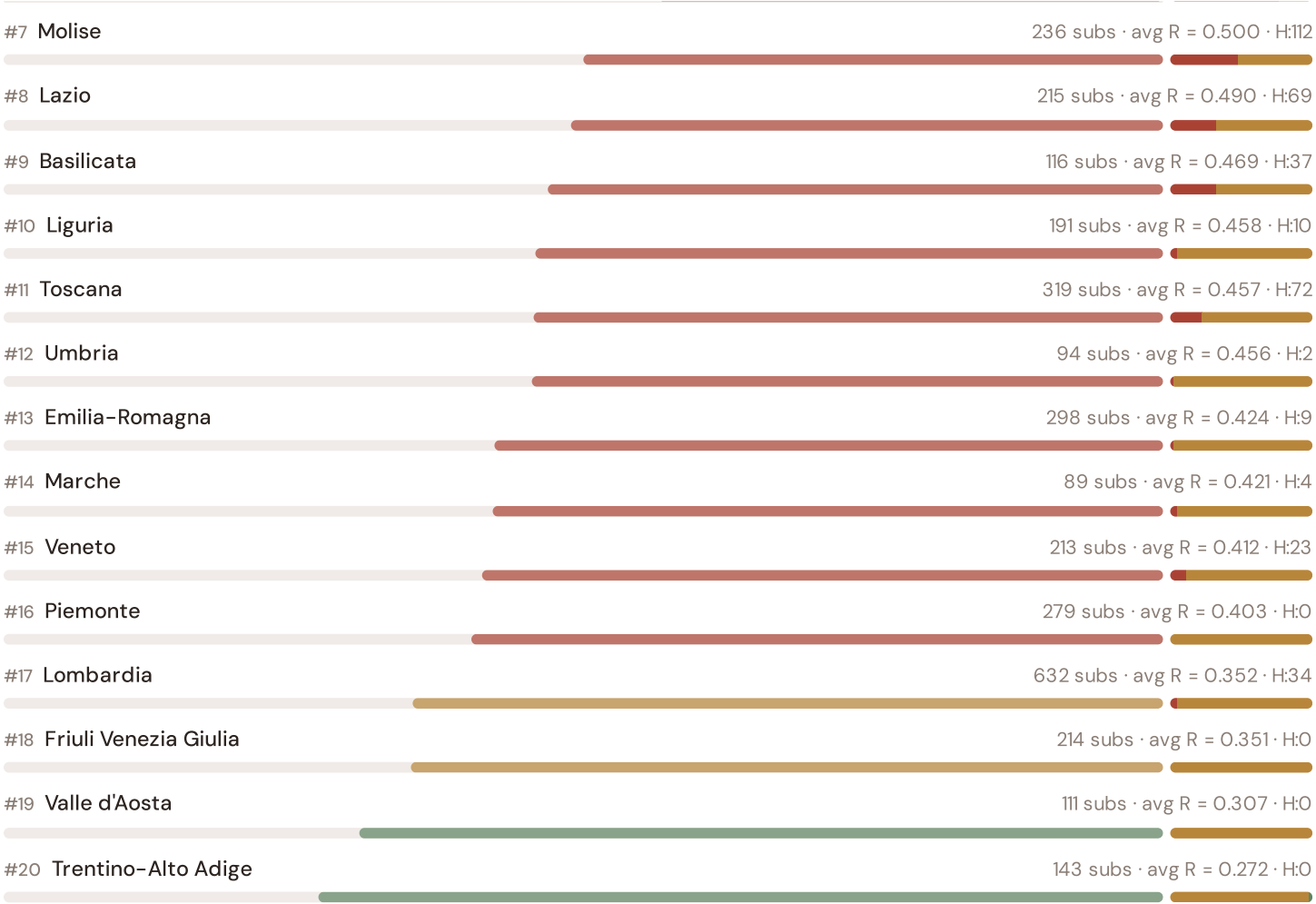
This month's key insight: The Calabria corridor deep-dive shows **106 substations** (of 106) with C-component above 70% of its weight ceiling — indicating continuity-of-supply performance consistent with rural territory classification. The corridor's average C of 0.755 is **2.1× the fleet average** of 0.362. The SSL reveals what aggregate national statistics conceal: that the country's mid-tier European position is not a uniform condition but a statistical average of urban-quality performance and rural-tier performance — and the corridor concentrates the worst of the latter.

Regional Risk Ranking — All 20 Italian Regions

Where does Puglia sit within the national landscape? Regions sorted by mean R_{median}, with band distribution and fleet comparison.

Longer bar = lower R = better resilience. Sorted worst → best.





The Calabria ranks **#1 of 20 region regions** by mean R_{median}, placing it among the upper tier of national risk. The three most stressed are Calabria, Sicilia, Campania, while the three most resilient are Trentino–Alto Adige, Valle d'Aosta, Friuli Venezia Giulia. 7 of 20 score above the fleet average of 0.491. This ranking — possible only because the SSI scores every substation individually rather than relying on aggregate DSO or national statistics — reveals the true spatial distribution of grid stress. It is precisely this substation-level granularity that positions the SSI as a tool for investment prioritisation: not treating entire regions as monoliths, but identifying the specific corridors within each that demand attention.

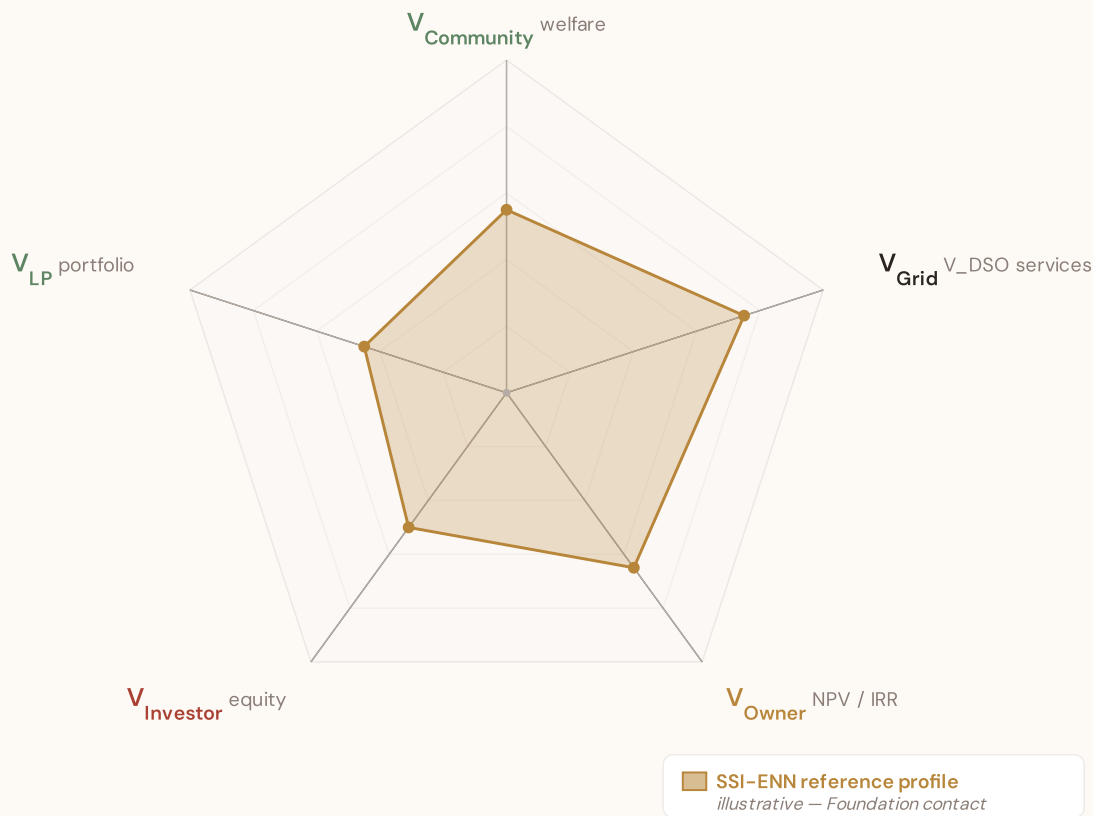
SECTION F — RECURRING MODULE

SSI Enhanced Neural Network — Monthly Topic

From grid intelligence to BESS valuation platform. The SSI Index answers *"How healthy is this substation?"* The SSI Enhanced Neural Network (SSI-ENN) extends this into: *"What is a BESS worth at this substation — to the DNO and to the community?"* Each edition of this report explores one building block of the SSI-ENN's five-layer architecture, showing how the SSI Index data feeds directly into investment-grade grid analytics.

SSI-ENN value-articulation radar — 5-lens stakeholder taxonomy

The SSI-ENN's commercial-tier value articulation answers "what is a BESS worth at this substation — and to whom?" through five stakeholder lenses (architecture memo §5 Decision #12): **V_{Community}** (welfare avoided outages, energy-poverty relief, employment) · **V_{Grid}** (V_DSO — avoided reinforcement, reliability, voltage, congestion, frequency, black-start, hosting capacity) · **V_{Owner}** (NPV / IRR / payback per asset class) · **V_{Investor}** (risk-adjusted equity return at the financing tier) · **V_{LP}** (limited-partner cash-on-cash + DPI / TVPI portfolio metrics). The pentagon shape below is an editorial reference profile, not per-substation commercial data — per-substation evaluation requires the SSI-ENN engines (L2-21 SupplyChainDD · L4-O5 Stakeholder Decomposition · L2-SYN Synergy). Foundation contact: ssi_index@ikenga.eu.



Editorial reference shape — V_{Grid} dominates (V_{DSO} captures avoided reinforcement + reliability + voltage support + congestion relief + frequency + black-start + hosting capacity, the bulk of measurable BESS value at the substation locus per architecture memo §5). $V_{Community}$ + V_{Owner} are co-secondary. $V_{Investor}$ + V_{LP} reflect the financing-tier waterfall above V_{Owner} NPV. Per-substation commercial decomposition lives in the SSI-ENN — mail to ssi_index@ikenga.eu for analytics.

Layer 1 — Data

SSI v4.0 component taxonomy, normalisation, MC engine

Layer 2 — Valuation

DSO/TSO + Community value stacks, TCO, real options

Layer 3 — Neural Net

GBT+DNN ensemble, transfer learning, SHAP

Layer 4 — Coverage

Country prioritisation, DCI, registry, calibration

Layer 5 — Platform

Products, API, revenue model, unit economics

LAYER 3

§0.5.5 · July 2026

Uncertainty Quantification

Investment-grade predictions require calibrated confidence intervals, not just point estimates

A point estimate without uncertainty bounds is unusable for investment-grade analysis. The SSI-ENN produces calibrated P10/P50/P90 prediction intervals using two complementary approaches: quantile regression (pinball loss) and conformal

prediction (distribution-free guaranteed coverage). The SSI's own Monte Carlo confidence intervals (CI_width, P5/P95) serve as the ground truth for calibration. This calibration is verified per country, per voltage class, and per risk band.

KEY CONCEPTS

→

Quantile regression (pinball loss) + conformal prediction

→

Calibrated against SSI Monte Carlo ground truth

→

Per-country, per-voltage, per-band calibration verification

→

Investment-grade: P10/P50/P90 for all outputs

LIVE SSI DATA CONNECTION

The training set comprises 4,293 substations with complete feature vectors. Average CI_width of 0.138 across the fleet provides the uncertainty ground truth that the neural network's quantile regression heads must calibrate against. The fleet's risk distribution — 4 Low, 2853 Medium, 982 High, 454 Critical — ensures balanced training across all risk bands.

Coming next month:

LAYER 2

Total Cost of Ownership Model — *Stochastic cost modelling — because BESS economics are not deterministic*

SECTION F.5 — V4.2 ANTI-MALADAPTATION GOVERNANCE GATE

Radar 2 — W1–W10 Substation Community Value

Wall 1 — Public W1–W10 framework.

The 10-axis Radar 2 expresses per-substation community-value baseline pre-intervention: *W1 Reliability · W2 Carbon · W3 Local Economic · W4 Climate Resilience · W5 Digital Inclusion · W6 Sovereignty · W7 Local Value Retention · W8 Communications · W9 Supply Security · W10 Compute Multiplier*. The discipline floor that the anti-maladaptation governance gate protects: *"no W axis worse off after intervention."*

5/1/1/3 4-tier audit-state classification (mesh resolution per Convention #6 §5.6)

Published (5) — solid fill

W1 · W2 · W3 · W4 · W8

Peer-reviewed methodology operational at fleet scale.

Baseline-only (1) — striped fill

W5 — Digital Inclusion

Foundation methodology under development; baseline value defensible at v4.2.

Published-baseline (1) ▲ — solid fill + chevron corner marker

W9 — Supply Security

Peer-reviewed academic basis (Albert/Holme topology-only baseline) with acknowledged scope limit (Buldyrev/IEEE 493 — N-1 contingency dynamics not captured at baseline).

Commercial (3) — dashed-faded

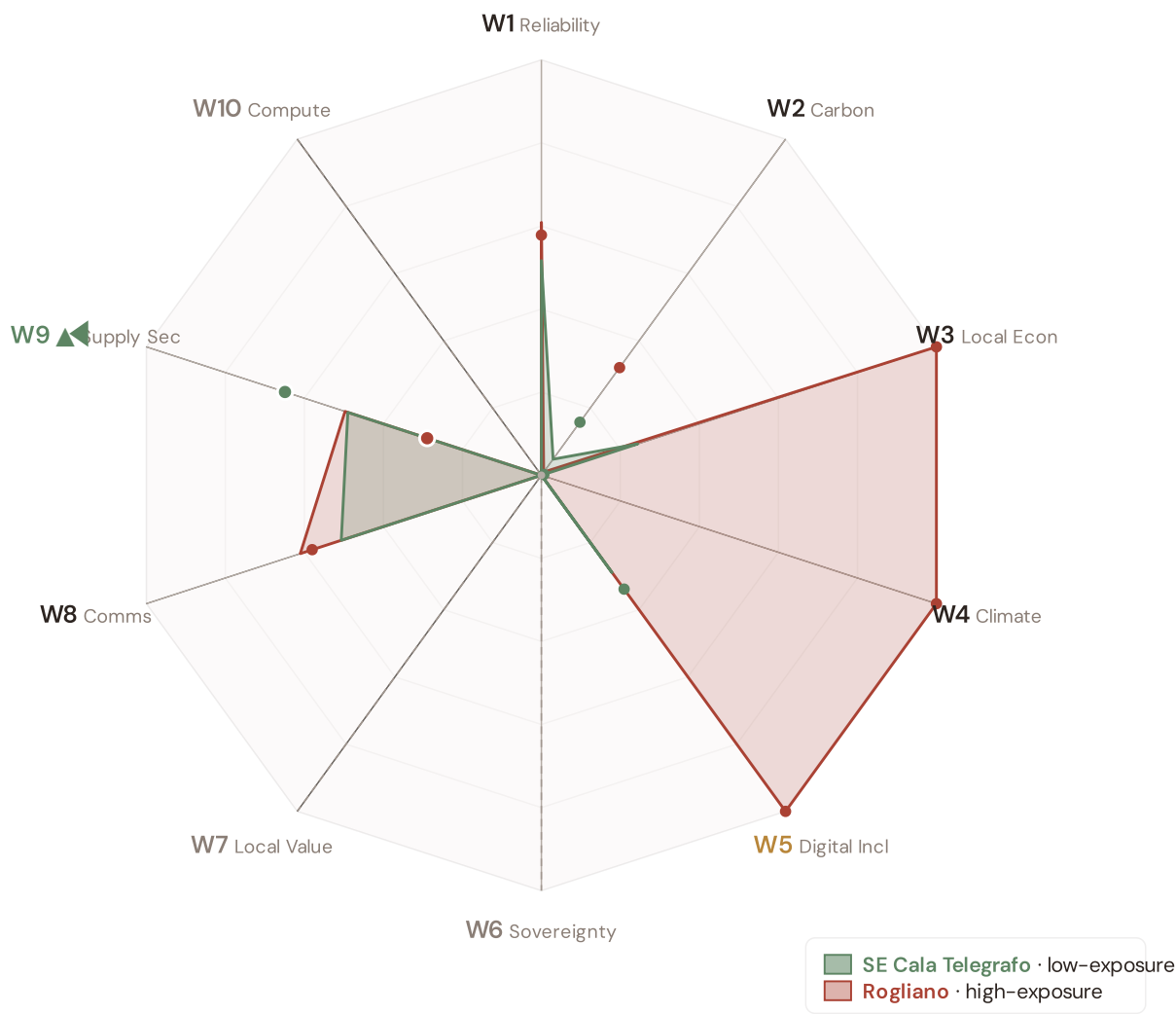
W6 · W7 · W10

Commercial deliverables — please contact the Foundation.

Per Convention #6 §5.6.7 academic-defensibility: the 4-tier classification aligns with IPCC AR6 + JRC COIN + Reckien (2021) continuum literature. The Published-baseline tier (W9) is the principled mesh resolution that honors both peer-review status + acknowledged scope limit without forcing a false binary.

Radar 2 visualization — anti-maladaptation governance gate

Two worked examples overlaid: **SE Cala Telegrafo** (low-exposure reference, Re_norm 0.170) and **Rogliano** (high-exposure reference, Re_norm 0.643). Axis labels are colour-coded per the 5/1/1/3 mesh resolution; the ▲ chevron marks W9 as Published-baseline. Hover any axis label for the methodology tooltip.



SE Cala Telegrafo polygon (sage): the low-exposure reference — compact footprint reflects this country's fleet floor on the W-axis baselines. Rogliano polygon (terracotta): the high-exposure reference — larger footprint where this country's fleet's W-axis values reach the upper percentile. Commercial-tier axes (W6 data sovereignty, W7 local value retention, W10 compute multiplier) display at zero per Convention #6 §5.6 mesh design — no public-data computation. W2 carbon uses Italy-specific zonal MEF per ssi_enn_v30 v4.2 calibration.

W1–W10 axis enumeration with audit–state classification

5/1/1/3 4-TIER · V4.2 MESH RESOLUTION · LIVE VALUES VIA COMPUTE_W_BASELINES.PY

AXIS	NAME	AUDIT STATE	METHODOLOGY BASIS
W1	Reliability	Published	$C \times E2_local \times population$ (ARERA SAIDI + ISTAT + ISTAT-POV)
W2	Carbon	Published	$T1 \times zonal\ MEF \times (V \cdot deg / 100)$ (Terna + EEA)
W3	Local Economic	Published	$norm(R10_just) - EU-SILC + OIPE\ energy-poverty$
W4	Climate Resilience	Published	$(R6d - 1) + (R6e - 1) + (R6c - 1) + \frac{1}{2}(R9 - 1) - v4.2$ resilience modifier family
W5	Digital Inclusion	Baseline-only	$norm(R8_adapt) - ARERA\ bulk\ SM + ENEA + DESI$
W6	Sovereignty	Commercial	Foundation contact (ssi_index@ikenga.eu)
W7	Local Value Retention	Commercial	Foundation contact (ssi_index@ikenga.eu)
W8	Communications	Published	$C \times p_crit_10yr \times population$ (Markov 10-yr critical probability)
W9 ▲	Supply Security	Published–baseline ▲	$0.60 \times BC_pct + 0.30 \times bridge + 0.10 \times (1 - deg / 20)$ — OSM power-graph betweenness centrality + bridge identification + degree-based redundancy
W10	Compute Multiplier	Commercial	Foundation contact (ssi_index@ikenga.eu)

▲ W9 Published–baseline — mandatory transparency tooltip

W9 — Supply Security (Published–baseline). Public-tier methodology: betweenness centrality + bridge identification + degree-based redundancy, computed deterministically from OpenStreetMap power-graph data. Peer-reviewed academic basis: Albert et al. (2000) *Nature*; Holme et al. (2002); broad power-grid centrality literature. Acknowledged scope limit (per Buldyrev et al. (2010) *Nature* + IEEE 493): topology-only baseline does not capture N-1 contingency dynamics or interdependent-network cascades. Richer methodology — N-1 contingency simulation + flexibility-services valuation — lives in SSI-ENN L2-22 / L2-23 commercial deliverables. Mail to ssi_index@ikenga.eu for extended analysis.

Worked examples. The W1-W10 per-substation baseline values for two anchor cases — SE Cala Telegrafo (low-exposure reference) and Rogliano (high-exposure reference) — are computed live by `_audit-snapshot/scripts/d3_radar2_per_country.py` against the country's `ssi-data.json`. Per Convention #6

§2.3.6 Path B refactor (11 Jun 2026), this country's exemplars are now algorithmically selected (min/max Re_norm) rather than named picks. See the FC v2 reference for W-axis methodology + commercial-tier rationale.

Anti-maladaptation discipline. The 10-axis Radar 2 baseline is the per-substation floor that any post-intervention scenario must protect. No W axis may degrade after a BESS deployment, line upgrade, or operational change. The discipline is enforced per axis (not in aggregate), and per substation (not in fleet average). This is the v4.2 governance gate that ensures grid-modernisation projects deliver net community value without hidden trade-offs.

SECTION G

Looking Ahead

NEXT MONTH — EDITION 03

Capital Metropolitan

Deep-dive into Region 02's grid risk profile — substation map, component analysis, region regions breakdown. European Context Card: Capital-intensive tier + flood overlay.

NEXT DATA REFRESH

Q4 2026 — 2 Quarterly Sources

GSE, Bpiitaly are due for refresh in Q4. Impact: DER registry (T1), business fabric indicators (E2), macro-economic data. Highest-impact annual source pending: **ISTAT — Italian National Statistics** (9 variables → R3 population, elderly, GDP, Mezzogiorno North-South split, fiscal capacity).

FLEET SPOTLIGHT

321 Bridge Substations — Single Points of Failure

321 substations are topological bridges — their failure disconnects part of the network. Of these, **94** are classified High or Critical. These are the fleet's single points of failure: high graph centrality (R4) combined with elevated risk makes them priority candidates for both grid reinforcement and BESS deployment.

Edition 03 will be published on the second Thursday of August 2026. Deep-dive region: **Region 02**. To ensure you receive it, confirm your subscription segment (General / Institutional / Academic) by email to ssi_index@ikenga.eu.

Feedback welcome. The SSI Index is designed to evolve through stakeholder dialogue. If you represent an Italian utility, ARERA, municipal energy company, or academic institution and would like to contribute data or methodology feedback, please contact us at ssi_index@ikenga.eu. The SSI's credibility depends on open scrutiny.

SSI Monthly Intelligence Report — Edition 005

Published 9 July 2026 · SSI Index **v4.2** (refreshed June 2026) · 4,293 substations · 14,221 grid lines · 6 components + **Re composite** (Radar 17th axis, bounds [0.920, 1.787]) · 20 metrics · 11 modifiers (5 baseline + 6 v4.2 resilience family)

Foundation contact

The SSI Index is a non-profit foundation project, please mail to ssi_index@ikenga.eu for more details.